5.3 Graphing Radical Functions

Name _____ Period ____ Date _____

Algebra II 5.3 (30-minute plan) · Student sheet

I can…

- graph radical functions.
- find the domain and range of a radical function.
- describe and write transformations of radical functions.

Vocabulary · fill in the missing word

Radical function — a function with the _____ in a radicand, like $f(x) = \sqrt{x}$ or $f(x) = \sqrt[3]{x}$

Domain of a square root function — the _____ that make the radicand ≥ 0

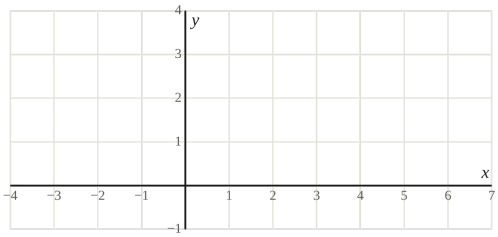
Transformations — $g(x) = a \cdot \sqrt[n]{x - h} + k$: a _____/shrinks (and reflects if $a < 0$), h moves left/right, k moves up/down

Part 1 · Notes and examples (fill in each note, then work the example)

Parent functions: $f(x) = \sqrt{x}$ has domain _____ and range _____. $f(x) = \sqrt[3]{x}$ has domain _____ and range _____.

Example 1

Graph $f(x) = \sqrt{x + 3}$. Identify the domain and range.



Transformations: a: In $g(x) = a \cdot \sqrt[n]{(x - h)} + k$: a stretches or shrinks the graph, and $a < 0$ _____ it in the x-axis.

Transformations: h and k: h translates the graph _____ ($h > 0$) or _____ ($h < 0$). k translates it _____ ($k > 0$) or _____ ($k < 0$).

Example 2

Describe the transformation of $f(x) = \sqrt{x}$ represented by $g(x) = -\sqrt{x - 4} + 1$.

Domain of a square root function: To find the domain of a square root function, set the radicand _____.

Example 3

Find the domain and range of $f(x) = \sqrt{2x - 6} + 1$.

Part 2 · Partner practice

Partner practice

Describe the transformation of $f(x) = \sqrt[3]{x}$ represented by $g(x) = \frac{1}{2}\sqrt[3]{(x + 5)} - 3$.